

Grade 11 Mathematics

Exponents and Surds & Equations and Inequalities

Task: Task 8

Term: Term 3 2026

Time: 60 minutes

Total Marks: 50

Examiner: AE Killer

Moderator: D Piters

QUESTION 1

- 1.1 Without using a calculator, verify each of the following. Show all your working.

1.1.1 Verify that $\sqrt{50} - \sqrt{18} = 2\sqrt{2}$ (2)

1.1.2 Verify that $(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3}) = 2$ (3)

- 1.2 Simplify each of the following without using a calculator. Show ALL working.

1.2.1
$$\frac{\sqrt[3]{8^2} \cdot 81^{\frac{3}{4}}}{\sqrt{144} \cdot 9^{\frac{-1}{2}}}$$
 (4)

1.2.2
$$\sqrt[5]{32^3} \cdot 32^{\frac{-4}{5}}$$
 (2)

1.3 Solve for x, giving all real solutions: $5\sqrt[3]{\frac{1}{x^2}} - 3 = 17$ (4)

- 1.4 Consider the surd expressions below.

1.4.1 Write $\frac{14}{3 - \sqrt{2}}$ with a rational denominator, in simplest form. (2)

1.4.2 Determine the rational numbers a and b if $(3 - \sqrt{2})(a + b\sqrt{2}) = 2 + 11\sqrt{2}$. (4)

- 1.5 In triangle ABC, the angle at B is 90° . The two shorter sides are $AB = \sqrt{3} + 1$ units and $BC = \sqrt{3} - 1$ units. D is the point on AC such that $BD \perp AC$.

1.5.1 Calculate the area of triangle ABC. (1)

1.5.2 Hence, determine the length of BD in simplest surd form. (3)

[25]

QUESTION 2

2.1 The sum of two numbers is 17 and their product is 66. Let the numbers be x and y . Write down two equations in x and y , and solve them to find the two numbers. (5)

2.2 By letting $k = x^2 - 3x$, solve for x : $(x^2 - 3x)^2 - 2(x^2 - 3x) - 8 = 0$ (5)

2.3 Solve for x : $2x(x-4) \leq (x-4)(x+3)$ (5)

2.4 Given: $f(x) = x^2 - 2x - 15$

2.4.1 Factorise $x^2 - 2x - 15$ fully. (1)

2.4.2 Hence, determine the values of x for which $\frac{\sqrt{15+2x-x^2}}{x-1}$ is real and defined. (4)

2.5 Given the quadratic equation $x^2 + (k-3)x + 4 = 0$, where k is a real constant.

2.5.1 Determine the value(s) of k for which the equation has equal roots. (3)

2.5.2 Hence, determine the values of k for which the roots of the equation are non-real. (2)

[25]

TOTAL [50]

MARKING GUIDELINE

QUESTION 1		
1.1.1	$\text{LHS} = \sqrt{25 \cdot 2} - \sqrt{9 \cdot 2} = 5\sqrt{2} - 3\sqrt{2}$	✓ simplify both surds
	$\text{LHS} = 2\sqrt{2} = \text{RHS}$	✓ match target (2)
1.1.2	$\text{LHS} = 5 - \sqrt{15} + \sqrt{15} - 3$	✓ expand
	$= 5 - 3$	✓ simplify
	$\text{LHS} = 2 = \text{RHS}$	✓ match target (3)
1.2.1	$\frac{(2^3)^{\frac{2}{3}} \cdot (3^4)^{\frac{3}{4}}}{12 \cdot (3^2)^{\frac{-1}{2}}}$	✓ convert to prime bases
	$\frac{2^2 \cdot 3^3}{12 \cdot 3^{-1}}$	✓ simplify each term
	$\frac{4 \cdot 27}{12 \cdot \frac{1}{3}} = \frac{108}{4}$	✓ simplify further
	27	✓ answer (4)
1.2.2	$(2^5)^{\frac{3}{5}} \cdot (2^5)^{\frac{-4}{5}} = 2^3 \cdot 2^{-4}$	✓ convert and simplify terms
	$\frac{1}{2}$	✓ answer (2)

1.3	$\sqrt[3]{\frac{1}{x^2}} = 4$	✓ isolate the root
	$x^{\frac{-2}{3}} = 4$	✓ single negative fractional exponent
	$x = \pm 4^{\frac{-3}{2}}$	✓ raise to reciprocal power
	$x = \pm \frac{1}{8}$	✓ both answers (4)
1.4.1	$\frac{14}{3-\sqrt{2}} \cdot \frac{3+\sqrt{2}}{3+\sqrt{2}} = \frac{14(3+\sqrt{2})}{9-2}$	✓ multiply by conjugate
	$6+2\sqrt{2}$	✓ answer (2)
1.4.2	$a+b\sqrt{2} = \frac{2+11\sqrt{2}}{3-\sqrt{2}} \cdot \frac{3+\sqrt{2}}{3+\sqrt{2}}$	✓ isolate bracket and rationalise
	$= \frac{6+2\sqrt{2}+33\sqrt{2}+22}{7}$	✓ expand
	$= \frac{28+35\sqrt{2}}{7} = 4+5\sqrt{2}$	✓ simplify
	$a=4$ and $b=5$	✓ equate rational and irrational parts (4)
1.5.1	Area = $\frac{1}{2}(\sqrt{3}+1)(\sqrt{3}-1) = \frac{1}{2}(3-1)$, so Area = 1 square unit	✓ answer

		(1)
1.5.2	$AC^2 = (\sqrt{3}+1)^2 + (\sqrt{3}-1)^2$	✓ substitution into Pythagoras
	$AC^2 = 4 + 2\sqrt{3} + 4 - 2\sqrt{3} = 8$, so $AC = 2\sqrt{2}$	✓ expand and simplify
	$\frac{1}{2} \cdot 2\sqrt{2} \cdot BD = 1$, so $BD = \frac{\sqrt{2}}{2}$ units	✓ answer (3)

QUESTION 2		
2.1	$x + y = 17$ and $xy = 66$, so $y = 17 - x$	✓ setup
	$x(17 - x) = 66$	✓ substitution
	$x^2 - 17x + 66 = 0$	✓ standard form
	$(x - 6)(x - 11) = 0$	
	The numbers are 6 and 11	✓ ✓ both answers (5)
2.2	$k^2 - 2k - 8 = 0$	✓ substitution, standard form
	$(k - 4)(k + 2) = 0$	✓ factors
	$x^2 - 3x = 4$ or $x^2 - 3x = -2$	✓ back-substitution
	$(x - 4)(x + 1) = 0$ or $(x - 1)(x - 2) = 0$	✓ factors

	$x = -1$ or $x = 1$ or $x = 2$ or $x = 4$	✓ answers (5)
2.3	$2x(x-4) - (x-4)(x+3) \leq 0$	✓ all terms on one side
	$(x-4)[2x - (x+3)] \leq 0$	✓ common factor
	$(x-4)(x-3) \leq 0$	✓ factors
	$3 \leq x \leq 4$	✓ ✓ answer, both parts (5)
2.4.1	$(x-5)(x+3)$	✓ factors (1)
2.4.2	$15 + 2x - x^2 \geq 0$	✓ setup inequality
	$x^2 - 2x - 15 \leq 0$, so $(x-5)(x+3) \leq 0$	✓ factors
	Critical values $x = -3$ and $x = 5$, giving $-3 \leq x \leq 5$	✓ interval
	$-3 \leq x \leq 5$ and $x \neq 1$	✓ exclude $x = 1$ (4)
2.5.1	$\Delta = (k-3)^2 - 4(1)(4)$	✓ substitution into discriminant
	$k^2 - 6k - 7 = 0$	✓ $\Delta = 0$ simplified
	$(k-7)(k+1) = 0$, so $k = 7$ or $k = -1$	✓ answer (3)

2.5.2	$k^2 - 6k - 7 < 0$, i.e. $(k-7)(k+1) < 0$	✓ condition $\Delta < 0$
	$-1 < k < 7$	✓ answer (2)

TOTAL [50]

COGNITIVE LEVEL ANALYSIS GRID

Weighting (Prescribed)			20%			35%			30%			15%			100%
Actual %			0%			56%			28%			16%			100%
Topic	Q	Sub-topic	Knowledge			Routine			Complex			Problem Solving			Total
			E	M	D	E	M	D	E	M	D	E	M	D	
Exponents and Surds	1.1.1	verify surd sum by simplifying				2									2
	1.1.2	verify product of conjugate surds				3									3
	1.2.1	surds with fractional exponents					4								4
	1.2.2	fifth root, negative exponent				2									2
	1.3	solve rational-exponent root equation									4				4
	1.4.1	rationalise the denominator				2									2
	1.4.2	equate rational, irrational parts											4		4
	1.5.1	area of triangle, surd sides				1									1
	1.5.2	pythagoras and area combined									3				3
Equations and Inequalities	2.1	sum-product word problem, quadratic				5									5
	2.2	substitution to reduce to quadratic					5								5

	2.3	quadratic inequality, common factor								5					5
	2.4.1	factorise quadratic trinomial				1									1
	2.4.2	surd domain, quadratic inequality												4	4
	2.5.1	equal roots, find parameter				3									3
	2.5.2	non-real roots, quadratic inequality								2					2
TOTAL			0	0	0	19	9	0	0	7	7	0	4	4	50
%			0%			56%			28%			16%			100%

E = Easy, M = Medium, D = Difficult.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$A = P(1 + ni)$$

$$A = P(1 - ni)$$

$$A = P(1 - i)^n$$

$$A = P(1 + i)^n$$

$$T_n = a + (n-1)d$$

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(r^n - 1)}{r - 1} ; r \neq 1$$

$$S_\infty = \frac{a}{1 - r} ; -1 < r < 1$$

$$F = \frac{x[(1+i)^n - 1]}{i}$$

$$P = \frac{x[1 - (1+i)^{-n}]}{i}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$M\left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2}\right)$$

$$y = mx + c$$

$$y - y_1 = m(x - x_1)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \tan \theta$$

$$(x-a)^2 + (y-b)^2 = r^2$$

$$\text{In } \triangle ABC: \quad \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

$$\text{area } \triangle ABC = \frac{1}{2}ab \cdot \sin C$$

$$\sin(\alpha + \beta) = \sin \alpha \cdot \cos \beta + \cos \alpha \cdot \sin \beta$$

$$\sin(\alpha - \beta) = \sin \alpha \cdot \cos \beta - \cos \alpha \cdot \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cdot \cos \beta - \sin \alpha \cdot \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cdot \cos \beta + \sin \alpha \cdot \sin \beta$$

$$\cos 2\alpha = \begin{cases} \cos^2 \alpha - \sin^2 \alpha \\ 1 - 2 \sin^2 \alpha \\ 2 \cos^2 \alpha - 1 \end{cases}$$

$$\sin 2\alpha = 2 \sin \alpha \cdot \cos \alpha$$

$$\bar{x} = \frac{\sum x}{n}$$

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$\hat{y} = a + bx$$

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$